Shuailong Dai

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WORKING EXPERIENCE

03/2024 – Present **National Energy System Operator (NESO)** **Senior Engineer**

- Developed and implemented an advanced Impedance Scan Toolbox to identify and analyze control interactions between Inverter-Based Resources (IBRs) and the grid, enabling precise assessment of small-signal stability in high-renewable penetration systems and supporting reliable integration of offshore wind, solar, and other low-inertia technologies critical to GB's net-zero ambitions.
- Led multidisciplinary investigations into Sub-Synchronous Oscillations (SSO) events on the GB transmission network, identifying root causes, conducting detailed Electromagnetic Transient (EMT) simulations, and delivering actionable technical recommendations to Original Equipment Manufacturers (OEMs)—directly enhancing system damping, preventing instability risks, and safeguarding secure, safe real-time operation of the national electricity system amid rapid decarbonization.
- Led multiple Ofgem-funded Network Innovation Allowance (NIA) projects focused on power system innovation, including tools for automated SSO identification, frequency-domain analysis, and control interaction studies—delivering measurable advancements in grid resilience, accelerated renewable connections, and consumer benefits through cost-effective, scalable solutions aligned with Great Britain's net-zero targets and energy security goals.

06/2024 – Present **International Council on Large Electric Systems (CIGRE) Working Group B4/C4.103 AC Network Equivalents for HVDC and FACTS Project Studies** **Specialist**

- Serve as a selected specialist expert in an international CIGRE Joint Working Group developing authoritative guidelines for creating accurate AC network equivalents in Electromagnetic Transient (EMT) simulation tools, specifically tailored for HVDC transmission systems and FACTS device project studies.
- Contribute specialist knowledge to address key challenges in modelling reduced-order AC system representations, ensuring reliable transient stability, interaction, and performance assessments for large-scale HVDC interconnections and FACTS applications in evolving power grids with IBRs.
- Participate in collaborative global expert discussions, technical reviews, and deliverable development—directly supporting innovation in grid planning, stability enhancement, and decarbonization technologies critical to the energy transition.

01/2026 – Present **CIGRE UK Next Generation Network (NGN)** **Vice-Chair**

- Provide strategic leadership and support to the NGN Chair in guiding the committee's activities for early-career professionals and students in the power systems sector, ensuring alignment with CIGRE's global mission to advance sustainable electricity systems and innovation.
- Chair Steering Committee meetings in the Chair's absence and contribute to decision-making on budget planning, event organization, member engagement strategies, and representation of young professionals' interests within CIGRE UK and internationally.
- Drive initiatives to facilitate the transition of young engineers into the power industry through technical resources, networking opportunities, industry-academia collaboration, and promotion of CIGRE participation—fostering talent development in critical areas such as renewable energy integration, grid resilience, and decarbonization.

Project Experience

04/2026 – 10/2027 **Frequency dependent network equivalent (FDNE) (£500,000)** **Proejct Lead**

- Engaged with stakeholders regularly and set up a communication plan;
- Developed frequency domain equivalent network with inverter-based resources (IBR) in PSCAD;
- Data-driven methods for cluster and aggregation on multiple operating points;
- Assured the deliverables are completed on time, within budget, and to the satisfaction of stakeholders.

04/2026 – 4/2027 **Hybrid impedance scan for network with IBRs (£400,000)** **Proejct Lead**

- Led Ofgem-funded NIA project and developing hybrid impedance methods to ensure network stability.
- Directed multidisciplinary team, managed budget/timeline, and engaged key stakeholders.
- Delivered innovative tools/methods for oscillation identification, advancing UK net-zero goals.

06/2024 – 12/2025 **Neural BB I&II (£200,000 and £400,000)** **Proejct Manager**

- Managed this project which is to use machine learning to create a surrogate model from a" black box" model of an AC/DC converter and provide technical support.

02/2022 – 12/2023 **'D3' - Data-driven Network Dynamic Representation for Derisking the HVDC and Offshore Wind, sponsored by NESO/Ofgen (£300,000)** **PhD Researcher**

- Developed one automated impedance scan toolbox for small signal stability analysis;
- Data-driven methods for cluster and aggregation on multiple operating points;
- Assured the deliverables are completed on time, within budget, and to the satisfaction of stakeholders.

EDUCATION

09/2020 – 03/2024	**University of Birmingham**	**PhD in Electronic, Electrical and Systems Engineering**
	PhD Project Title:	Impedance modeling and stability analysis of grid-connected IBR
	Highlights of the PhD thesis:	Impedance scanning toolbox and transfer function estimation; Automated impedance-based stability analysis on control interaction; Data-driven cluster and aggregation on multiple operating points.
09/2018 - 06/2020	**China Three Gorges University**	**Master in Electrical Engineering**
	MPhil Project Title:	Power management and control of lithium battery packs
	Courses:	Power system operation and control; Modern power electronics technology;
09/2014 – 06/2018	**China Three Gorges University**	**Bachelor in Electrical Engineering and its Automation**
	Undergraduate Project Title:	Lithium battery SOC equalization circuit design for electric vehicles

PUBLICATIONS

1. Jay Ramachandran, Weihua Zhou, Charlie Guo, **Shuailong Dai**. "Benchmarking of Six Different Frequency Scan Tools – Insights and Lessons Learned" CIGRE Australia Conference 2025.
2. **Shuailong Dai,** "Frequency Scan Toolbox: Data-Driven Solution for Impedance-Based Stability Analysis and Network Representative", CIGRE UK B4 Technical Liaison Meeting 2025.
3. **Shuailong DAI**, Chengyi WU, David LI, Dechao KONG, Xiaoyao ZHOU, Xiao-Ping ZHANG*, "A Methodology to Derisk HVDC and Offshore Wind Connections to a Network", CIGRE Paris Session 2024.
4. **Shuailong DAI** and Xiao-Ping ZHANG, "Advanced Identification Methods for Power System Oscillations based on Measurements," 2022 IEEE 16th International Conference on Compatibility, Power Electronics, and Power Engineering (CPE-POWERENG), Birmingham, United Kingdom, 2022.
5. **Shuailong Dai**, Feiran Zhang, Xiao Zhao., "Series-Connected Battery Equalization System: A Systematic Review on Variables, Topologies, and Modular Methods" International Journal of Energy Research 45 (2021): 19709 – 19728, doi.org/10.1002/er.7053.
6. **Shuailong Dai,** Bin Cao, Rui Liu, Liyang Zhu, Mengfan Li, Power Balance Optimization Technology of Microgrid Based on Full‐Bridge Converter, Complexity, 2020.
7. **Shuailong Dai,** Jiayu Wang, Teng Li, Zhifei Shan, Yewen Wei Analysis, design and implementation of flexible interlaced converter for lithium battery active balancing in electric vehicles, Journal of Power Electronics, 2019.
8. **Shuailong. Dai,** Y. Wei, J. Wang, J. Li and Z. Ouyang, "Research on Hybrid Natural and Forced Active Balancing Control of Battery Packs State of Charge Based on Unscented Kalman Filter," IEEE International Power Electronics and Application Conference and Exposition (PEAC), 2018.

HONORS & AWARDS

2026	GELA Global Sustainability Youth Leaders (35 under 35)	
2022	IEEE PES University of Birmingham Student Paper Competition & Open Presentation Competition	Third Winner
2020	China National Postgraduate Electronics Design Contest	The first prize
2019	China National Mathematical Contest in Modeling	The first prize
2019	China National University Student Innovation & Entrepreneurship Development Program (£11,000)	Team Leader
2018	‘TGOOD’ Innovation and Entrepreneurship Seed Fund (£7,000)	Team Leader
2018	China International College Students' 'Internet+' Innovation and Entrepreneurship Competition	The third Prize

MEMBERSHIPS & VOLUNTEERS

2026 - Present	IET Circuits, Devices & Systems	Academic Editor
2026 - Present	IEEE Transactions on Power Systems	Reviewer
2022 - Present	International Council on Large Electric Systems (CIGRE) UK	NGN Member
2018 - Present	Institution of Engineering and Technology (IET)	Member
2016 - Present	Institute of Electrical and Electronics Engineers (IEEE)	Member

SKILLS

▪ Proficient in PSCAD, Python, Matlab, Scilab, PSIM, Altium Designer (PCB), Visio, Origin

▪ General knowledge of RTDS, PowerFactory, MongoDB (NoSQL), MySQL (SQL), RTDS, Opal-RT, R, NetLogo